

Fig. 11: Sample Ladder program

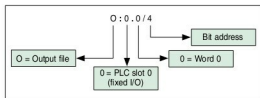


Fig. 12: A typical address for AB PLC

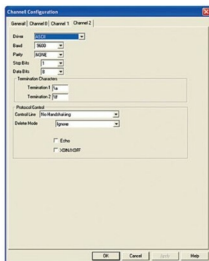


Fig. 13: Setting RS232 channel configuration

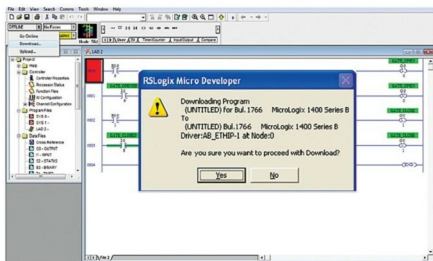


Fig. 14: Downloading the Ladder program to the PLC

An Example of BIT Addresses and Their Functions in the PLC

BIT address	Button (function)
CS:1.ACC	Current cycle status
B3:0/10	Start
B3:4/1	Acid pump
I:0/3	Fume hood
B3:4/0	Chiller
T4:2.TT	PS status
N7:0	Acid temp
CS:1.pre	Total cycle
B3:0/0	Verbose cmd read
B3:0/1	Cmd read2
B3:0/2	Max current
B3:0/3	18V
B3:0/4	Read current
B3:0/5	Cmd read5
B3:0/6	OV power supply via rs232
B3:0/7	Read MV
B3:0/8	Cmdread8

vertical lines to represent input power rails and stringing the rest of the circuit between these. Sub-routines are implemented as another ladder.

A typical example of a block used for RS232 programming is AWA (Fig. 9).

ASCII write append (AWA). The AWA instruction sends a specified number of characters (for example, 50) of the source tag (ST14:0) to a PLC serial port 2 and appends either one or two pre-defined characters. It is used for RS232 communication.

Similarly, an SCL is an instruction used to read and scale an analogue input to a corresponding physical value like temperature or pressure.

In the above example, the address for the analogue input is I: 1.0 and the converted raw data is divided by 10,000 and added to offset to implement a linear transfer function like

$$y = mx + c$$

A number of free Ladder programming tutorials are available on the Internet. Inside the PLC, we have various registers like output, input, status and binary. Each bit in these registers specifies a function. Addresses of the devices are built on the basis of names, PLC slot numbers, words and bit numbers. Address representations differ from manufacturer to manufacturer.

The table above shows some of the bit addresses of the PLC and how these are related to field devices.

A typical address of an AB PLC for output coil is built as in Fig. 12.

Once you write your Ladder code for the PLC, download the code into the PLC. Let the PLC be in Run mode after that.

Fig. 13 shows how an RS232 port can be configured for handshaking a power supply device, which has settings 9600, N, 8, 1.

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